

Filotas Theodosiou

Senior Data Scientist | Forecaster

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Summary

Senior applied researcher and lecturer specializing in ML-based time series forecasting for complex business settings. Member of the International Institute of Forecasters (IIF) and the Early Career Forecasters (ECF). Designs and deploys end-to-end forecasting systems, from scoping and data pipelines to modeling, MLOps, and decision-support delivery. Maintains active research and contributes to open-source. Regular presenter at forecasting conferences, with a critical focus on practical AI/LLM capabilities and limitations.

Core Skills

- **Problem-Solving:** Scoping, solution architecture, stakeholder alignment, clear communication
- **Forecasting:** Large Scale Operational Forecasting, Global models, hierarchical reconciliation, uncertainty estimation, Gradient Boosted Decision Trees,
- **Tech:** Advanced Python; fluent in R; ML expertise; MLOps; AWS; Google Cloud
- **AI Tools:** Deep understanding of LLM tooling (text/coding), experience with agentic AI apps

Education

- **BSc, Mathematics** — Aristotle University of Thessaloniki
- **MSc, Data Science** — University of Skövde (Sweden)

Selected Projects

Food Waste Reduction via Enhanced Demand Forecasting (2024-Present)

Designed and deployed forecasting systems for ready-to-eat sector partners; delivered 13%-133% forecasting accuracy improvements across five partners, up to 67% food-waste reduction per meal, and 4%-20% estimated operational cost reductions. Presented at ISF 2025.

End-to-End Forecasting System: Architecture and Deployment (2025-Present)

Designed an event-driven forecasting platform covering data pipelines, forecasting, probabilistic decision support, and optimization workflows; currently migrating components toward on-premise architecture to ensure continued client support.

Sales Forecasting for SMEs with Limited Data (2022-2024)

Global modeling plus hierarchical reconciliation to overcome data scarcity of SMEs; 40% Accuracy Improvement over previous approaches. Presented at ISF 2024.

AI Education: Early Risk Detection and Interventions (2022-2024)

Early risk detection for students based on online activity; SHAP-based interventions for study adjustments; Presented at ISF 2023.

Open Source

- **DeepRetail**: Forecasting package with reconciliation methods — [GitHub](#)
- **Grafton**: GDPR anonymizer (consent, encoding key, randomized numbers) — [GitHub](#)
- **Covid-19 Impact under Parametric Uncertainty** — [GitHub](#)
- **Real-Time Twitter Emotion Detection** — [GitHub](#)
- **Music Recommendation System** — [GitHub](#)

Selected Papers

- **2025** — Probabilistic Forecasts with Global GBDTs: Rethinking Bootstrapping with Structural Adjustments
- **2024** — Cross-Temporal Supervised Learning for Collaborative Inventory Management
- **2024** — Evaluation of Early Student Performance Prediction given Concept Drift
- **2023** — Cross-Temporal ML Forecasting with Limited Historical Demand Data
- **2021** — Forecasting with Deep Temporal Hierarchies
- **In progress** — The Limits of Scale: An Information-Based Framework for Time Series Forecasting

Selected Talks

- **2025** — Navigating the (Noisy) LLMs Landscape (ECF 2025, Beijing)
- **2025** — Integrating Smooth Capabilities into Python (Open Source Forecasting Workshop, Beijing)
- **2025** — Probabilistic Forecasts with Global GBDTs (ISF 2025, Beijing)
- **2024** — Probabilistic Forecasts for Global Models (ISF 2024, Dijon)
- **2024** — How AI is Applied to Reduce Food Waste (Internal Research Seminar, Kortrijk)
- **2023** — Forecasting with Temporal Hierarchies (Short Research Workshop, Kortrijk)
- **2023** — Modern Practices in Business Forecasting: GBDTs (Forecasting Applied Workshop, Kortrijk)
- **2021** — Forecasting with Deep Temporal Hierarchies (ISF 2021; Research Seminar; ICAIF 2021)

Teaching

- **2023-Present** — Course: Cloud for AI (MLOps) — VIVES University of Applied Sciences

All presentations and their slides plus the paper abstracts and course materials are available on the [website](#).